How Far Can We Trust a Deep Probabilistic Time Series Forecast?

Investigating Calibration Degradation in DeepAR and TFT, and its Post-hoc Conformal Correction via EnbPI

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***Abstract*—Deep probabilistic forecasting models such as DeepAR and the Temporal Fusion Transformer (TFT) are widely used for predictive distribution and quantile prediction for multi-horizon time series. Despite their usage in the field, the probabilistic calibrations are assessed in an aggregated way across all forecast horizons. This aggregate approach has a critical structural blind spot: a model that is well-calibrated at horizon h=1 may have miscalibration at h=96. Current existing benchmarks provide no methods to detect this degradation.**

This research identifies this *horizon-conditional calibration gap* as an evaluation limitation of the deep probabilistic forecasting literature. We propose a horizon-stratified evaluation framework that decomposes the conditional coverage rate, Continuous Ranked Probability Score (CRPS), Probability Integral Transform (PIT) uniformity, interval width, quantile-crossing rate, and Weighted Interval Score (WIS) across forecast horizons. This framework quantifies the degree of calibration degradation in DeepAR and TFT across three benchmark time-series datasets (UCI Electricity, UCI Traffic, and the M5 Walmart dataset). We further propose applying Ensemble Batch Prediction Intervals (EnbPI) [1] as a post-hoc horizon-adaptive conformal correction method to assess whether coverage guarantees can be restored. The expected contribution is a widely applicable, systematic evaluation protocol for probabilistic forecasting tasks that reveals previously unmeasured failures, with practical value for forecasting in energy, logistics, and retail demand management.

***Index Terms*—probabilistic forecasting, uncertainty quantification, calibration, prediction intervals, conformal prediction, time series, deep learning**

I. Introduction

When forecasts are used to make real decisions, knowing how uncertain a prediction is as critical as the prediction itself. In energy grid management, a prediction that underestimates demand can result in a risk of reserve capacity short [2]. In traffic network management, poorly generated occupancy forecasts result in inefficient signal control and under-utilisation of infrastructure [3]. In retail, an under-confident prediction may lead to a persistent understocking [4]. These examples show why *point forecasting* shifted to *probabilistic forecasting*, which covers the uncertainty of predictions, which is not just a mere theoretical exercise, but a practical requirement for each field.

The past literature has shown gradual progress in deep probabilistic forecasting. DeepAR [5], which is a built-in algorithm in Amazon SageMaker AI [6], is based on the iterated approach. A single LSTM is trained on thousands of related series rather than fitting the model in a separate model per series. DeepAR generates autoregressive one-step-ahead Gaussian predictions, with multi-step forecasts by recursively inputting each prediction as the next step's input [5]. TFT [7] advances DeepAR's strategy by adopting a direct sequence-to-sequence architecture, eliminating autoregressive feedback. TFT is composed of a transformer-based decoder that produces all multi-step predictions in a single forward pass.

However, simply knowing how to use these models does not ensure the certainty of the predictions. How we should evaluate these models' probabilistic forecasts connects to how strictly proper scoring rules [8] and the calibration-sharpness decomposition [9] have evolved into the current evaluation protocols as the theoretical foundation. Yet a main evaluation question remains unanswered: *do these models guarantee calibrations as the forecast horizon increases?*

Across the existing literature, from deep model evaluations [5], [7] and forecast scoring studies [10] to benchmarking such as GluonTS [11] and the M5 Uncertainty Competition [12], evaluation protocols consistently show metrics such as Scaled Pinball Loss (SPL) and Weighted Interval Score (WIS) that are aggregated across all forecast horizons. No existing framework provides a mechanism to detect whether calibration degrades at longer horizons.

Leaving this problem unresolved is not merely an academic issue. A practitioner deploying DeepAR for 24-step-ahead electricity demand forecasting receives aggregate validation metrics that look satisfactory, while the model's 90% intervals at h=20 may actually cover only 60%, leading to risk in the practical operation.

For DeepAR, the autoregressive Gaussian assumption structurally breaks down as h grows [5]. For TFT, independent optimisation of quantile heads may result quantile-crossing, where a predicted 90% interval falls below the 50% interval, which is also likely to be driven by structurally long horizons [7].

Even with a clear risk of failure in these deep probabilistic forecasting models, no evaluation tool exists to examine per horizon. Thus, this research proposes to close this gap.

II. Critical Literature Review

A. Cluster A: Foundations of Probabilistic Methods

The basis of evaluating probabilistic forecasts relies on Gneiting and Raftery [8] and Gneiting et al. [9].

Gneiting and Raftery [8] define a scoring rule as *strictly proper* only if forecaster report their true predictive distribution. They prove that the CRPS can be applied as a strictly proper scoring rule that which is also sensitive to distance. CRPS measures the area difference between the predicted Cumulative Distribution Function (CDF) and the step function that places all mass a realised value y. Thus, a forecast distribution concentrated near y yields a lower score. Any alternative point-forecast metrics such as RMSE or MAE discards distributional information entirely and therefore provides no guarantee honest reporting [8]. However, the critical limitation remains within [8] as it introduces CRPS for a single forecast-observation pair which does not regard sequence of multi-step-ahead forecasts where residuals may serially correlated.

Gneiting et al. [9] define that forecasters should "maximise sharpness subject to calibration," where calibration requires that the Probability Integral Transform (PIT) applied to the predictive CDF follows a uniform distribution. However, this calibration is defined *marginally*. The PIT values are aggregated across all time steps and evaluated jointly. This method logic is motivated by the i.i.d. (Independent and Identically Distributed) weather ensemble setting, causing a structural limitation for multi-step forecasting. A model whose calibration that degrades gradually with horizon h may still produce a PIT that is approximately uniform when aggregated across h, hiding the degradation.

Bracher et al. [10] introduce the Weighted Interval Score (WIS), a proper scoring rule for interval prediction. WIS builds on the Interval Score (IS) for a single prediction interval at nominal level $(1 - \alpha_K)$, which penalises interval width and adds a $2/\alpha_K$-scaled penalty when observation is out of the interval. Then, WIS aggregates K IS values with weight $\frac{\alpha_K}{2}$ multiplied to each IS_K and then summed up together. Bracher et al. [10] confirm that $K = 11$ yields $\mathrm{corr}(\mathrm{WIS}, \mathrm{CRPS}) \approx 1.0$ as based on Riemann sum as K grows, WIS converges to CRPS. WIS decomposes into spread, underprediction, and overprediction components, enabling richer diagnostics than CRPS alone. This metric used by the COVID-19 Forecast Hub is closely related to the M5 Weighted Scaled Pinball Loss (WSPL) [12], both sharing the same Pinball Loss foundation.

As IS scores a single time point like CRPS, Bracher et al. [10] note that investigating scores separately by horizon is important, yet their empirical evaluation only provides the average of WIS across all forecast horizons. This way of inheriting the same horizon aggregated evaluation blind spot as CRPS and PIT, which fail to detect whether miscalibration exists in a specific horizon.

B. Cluster B: Classical Quantile Regression

Koenker and Bassett [13] introduce quantile regression by showing that minimising the tilted absolute loss $\rho_\tau(u) = u(\tau - \mathbf{1}[u < 0])$ where u is the residual, yields the conditional τ-quantile of the response distribution. This pinball loss function is applied by a quantile-head neural structure, including TFT [7]. However, these deep learning implementations use independent optimisation of each quantile level, missing a mechanism to ensure *quantile ordering*: $\hat{q}_{0.1}(x) \leq \hat{q}_{0.5}(x) \leq \hat{q}_{0.9}(x)$. Quantile crossing, where this ordering is broken, can be worsened at longer forecast horizons.

Moreover, Chung et al. [14] demonstrate that minimising pinball loss does not guarantee calibration. For any average-calibrated model, even with lower pinball loss and sharper prediciton, the model failed in calibration. This implies the necessity of evaluating the coverage rate and the prediction interval's width separately.

C. Cluster C: Deep Probabilistic Forecasting Models

DeepAR [5] trains a single LSTM autoregressive model across thousands of time series using negative log-likelihood. It outputs parameters of a Gaussian or Negative Binomial distribution per step. DeepAR demonstrates that global multi-series training outperforms series-local methods. However, DeepAR's true predictive distribution drifts gradually from Gaussian as the LSTM goes through sequential autoregressive steps. As uncertainty propagates through these steps, the accumulated predictive distribution deviates from the Gaussian assumption, causing calibration to degrade with horizon. Moreover, DeepAR's evaluation reports aggregate ρ-risk (quantile loss: the pinball loss averaged across quantile levels) and ND (Normalised Deviation: $\sum |\hat{y} - y| / \sum |y|$), providing no horizon-stratified calibrations.

TFT [7] advances this iterative model by multi-head self-attention over temporal features, gating networks for variable selection, and quantile regression heads at multiple levels. Yet TFT optimises each quantile head via an independent pinball loss, allowing quantile crossing. Additionally, attention mechanisms have implications that temporal attention may degrade with increasing horizon which is never been tested. Critically, the paper's evaluation is aggregate P50/P90 quantile loss, with no PIT histogram, coverage-by-horizon, interval width measure or quantile crossing analysis.

D. Cluster D: Diffusion Model in Generative Models

Rasul et al. [15] introduce TimeGrad, applying denoising diffusion probabilistic models (DDPMs) to multivariate time series. An RNN encodes temporal context, conditioning a DDPM that generates future values via a reverse diffusion chain. TimeGrad achieves considerable multivariate accuracy but requires 100+ denoising steps per sample, making it impractical for long-horizon prediction. Evaluation is on fixed-length horizons with aggregate CRPS.

Similarly, Tashiro et al. [16] propose CSDI, a conditional score-based diffusion model that learns the posterior distribution of missing values using a two-dimensional transformer

attention over temporal and feature dimensions. Forecasting is treated as a masked imputation. CSDI reports aggregate CRPS averaged over all imputation targets, showing that no per-horizon evaluation is confirmed.

E. Cluster E: Conformal Prediction for Time Series

Xu and Xie [1] introduce Ensemble Batch Prediction Intervals (EnbPI). With bootstrap ensembles and sliding-window residual updates, EnbPI relaxes the i.i.d. exchangeability requirement. At each prediction step, EnbPI gathers the most recent residuals in a sliding window to generate prediction intervals. Yet the paper's coverage guarantee is a *marginal* coverage rate averaged across all predictions regardless of horizons, and only shows guarantees for one-step-ahead predictions.

Gibbs and Candès [17] introduce conformal prediction for non-stationary settings through Adaptive Conformal Inference (ACI). It adjusts the miscoverage rate $\hat{\alpha}_t$ using a feedback loop on past coverage outcomes. However, the adaptation is applied over calendar time, not over forecast horizons, thus horizon-conditional coverage is not guaranteed or analysed.

F. Cluster F: Evaluation Infrastructure

GluonTS [11] and the M5 Uncertainty Competition [12] are reviewed together because they introduce one of the main evaluation infrastructures of the time series field. GluonTS provides standardised code toolkits for dataset splits, model implementations, and evaluation via aggregate CRPS, ND, and quantile loss, which miss the horizon-stratified analysis. The competition, which has the largest probabilistic forecasting benchmarks with ∼900 teams participating, uses WSPL aggregated across all 28 forecast days. The competition acknowledges that "calibration remains challenging" but again provides no horizon-stratified analysis. These show the gap is acknowledged in the field's largest empirical study, but never addressed.

G. Synthesis: Missing Horizon-Conditional Calibration

Across all six clusters, calibration is consistently evaluated in aggregate form, over all forecast horizons [11], [12]. This convention hides systematic degradation at specific horizons, especially for strong short-horizon performance, canceling out poor long-horizon miscalibration.

Probabilistic forecasting models mentioned in the Cluster C&D are evaluated on aggregated metrics. Conformal prediction wrappers (Cluster E) provide marginal, not conditional, coverage. Benchmarks (Cluster F) provide aggregate metrics as the standard. These gaps in methodological clusters show missing horizon-conditional calibration, which can measure how accurate it is at each specific horizon h.

III. Research Gap and Objectives

Gap Classification: Evaluation gap.

Gap Statement: Current deep probabilistic forecasting models and their evaluation frameworks assess calibration via metrics aggregated across forecast horizons. A systematic, horizon-stratified calibration analysis across model architectures is absent in previous studies. The measurement of *how* and *by how much* existing models fail across horizons remains missing. This gap thus motivates this research, which additionally applies EnbPI as a post-hoc conformal prediction wrapper to investigate whether horizon-conditional coverage can be restored after the failure profile is confirmed.

Research Question: *What is the maximum reliable forecast horizon h^* for deep probabilistic models (DeepAR, TFT) across structurally different dataset types, and can horizon-conditional conformal calibration extend h^*?*

Here $h^* = \min\{h : \text{Coverage@}h < 1 - \alpha - \varepsilon\}$ is the first horizon where observed coverage drops more than ε below the nominal level $1 - \alpha$. Metrics evaluated along with Coverage@h are defined in Table I. While h^* is derived from Coverage@h alone, CRPS@h and PIT Kolmogorov–Smirnov@h (PIT KS@h) are evaluated together for both models, and Crossing@h, Width@h, and WIS@h are measured only for TFT degradation patterns.

SMART Objectives:

O1 Evaluate horizon-conditional CRPS@h, Coverage@h, PIT KS@h for both DeepAR and TFT. For TFT only, measure Width@h, WIS@h. Evalaute both models on three benchmark datasets across $h = 1, \ldots, H$ (where $H \in \{24, 28\}$ per dataset). Examine both dataset-specific and model-specific h^*.

O2 Quantify quantile crossing rate dependent on h for TFT and identify structural causes (attention entropy decay, distribution mismatch).

O3 Apply EnbPI as a post-hoc horizon-adaptive conformal prediction wrapper to recalibrate DeepAR and TFT outputs. Assess: (i) how Coverage@h recovers to $1-\alpha$, (ii) Diebold-Mariano (DM) tests on CRPS@h (WIS@h for TFT) before vs. after recalibration at $p < 0.05$ with Bonferroni correction ($\alpha_{\text{adj}}=\alpha/H$), and (iii) examine horizons whether residual miscalibration remains (PIT KS@$h > c_\alpha$).

O4 Design a reusable horizon-stratified calibration evaluation protocol as open-source Python code integrated with GluonTS.

IV. Proposed Methodology

A. Technical Pipeline

The proposed pipeline consists of three phases, shown in Algorithm 1. Sections IV-B and IV-C explain model selection, metric definitions, and the statistical rationales for each phase.

B. Model and Baseline Justification

DeepAR [5] is selected as the probabilistic baseline because its Gaussian output assumption represents the simplest and most common distributional choice, making horizon-dependent miscalibration most visible.

At each timestep, DeepAR concatenates the scaled previous target value with time-varying covariates into a flaten input vector $\mathbf{x}_t = [y_{t-1}/\nu_i,\ \sin(2\pi t/T),\ \cos(2\pi t/T),\ \ldots]$, where ν_i is the per-series mean scale [5]. The LSTM hidden state is converted into the parameters (μ_t, σ_t) of a Gaussian (or Negative Binomial for count data) distribution. All covariates

Algorithm 1 Horizon-Stratified Calibration Evaluation

Require: $\mathcal{D}$, $M \in \{\text{DeepAR, TFT}\}$, H, T, $\mathcal{Q}$, B, $1-\alpha$, ε
Ensure: h^*_{pre}, h^*_{post}, Δh^*, profiles $\{\text{CRPS@}h, \text{Coverage@}h, \text{PIT KS@}h\}_{h=1}^{H}$ before/after

Split $\mathcal{D}$; train M (model-specific framework: see §IV-B)
Phase 1 – Baseline Calibration Profiling:
for $t = 1, \ldots, T$ **do**
  Generate $\hat{F}_{t,h}$ for all h; compute $\text{PIT}_{t,h} \leftarrow \hat{F}_{t,h}(y_{t+h})$
  Compute accuracy $\text{score}_{t,h}$: CRPS (DeepAR) or WIS+Crossing+Width (TFT)
end for
Aggregate metrics@$h \leftarrow \frac{1}{T}\sum_t(\cdot)_{t,h}$ for all metrics; compute PIT KS@h, Coverage@h
$h^*_{\text{pre}} \leftarrow \min\{h : \text{Coverage@}h < 1-\alpha-\varepsilon\}$
Phase 2 – Conformal Recalibration (EnbPI):
Train B LOO (Leave-One-Out) bootstrap models $\{M_b\}$; compute $\hat{\varepsilon}_{i,h} \leftarrow y_{i+h} - \hat{f}_{-i}(x_i, h)$
for each $h = 1, \ldots, H$ **do**
  Derive score s_h from $\{\hat{\varepsilon}_{i,h}\}$; adjust $\hat{C}^{\text{CP}}_{t,h} \leftarrow [\hat{q}_{\alpha,t,h} - s_h,\ \hat{q}_{1-\alpha,t,h} + s_h]$
  Update s_h online (sliding-window); recompute Coverage@h with $\hat{C}^{\text{CP}}_{t,h}$
end for
$h^*_{\text{post}} \leftarrow \min\{h : \text{Coverage}_{\text{CP}}\text{@}h < 1-\alpha-\varepsilon\}$
Phase 3 – Statistical Testing:
DM test on CRPS@h (WIS@h for TFT) pairs before/after; Bonferroni ($\alpha_{\text{adj}}=\alpha/H$)
Flag horizons where post-EnbPI PIT KS@$h > c_\alpha$
return h^*_{pre}, h^*_{post}, $\Delta h^* = h^*_{\text{post}} - h^*_{\text{pre}}$, calibration profiles before and after conformal recalibration

including static and time-varying are treated equally, giving no structural separation by input type. Moreover, DeepAR is *autoregressive*, which accumulates non-linearities throughout the prediction chain cause the predictive distribution to shift from Gaussian output form as h grows.

TFT [7] is selected for (i) its independent quantile optimisation allowing quantile crossing, and (ii) its attention mechanism that can provide an additional diagnostic axis (attention entropy vs. horizon).

Structurally, TFT separates its inputs into three types processed by distinctive encoders: *static*, *past observed* (historical target values and past covariates), and *known future* (future covariates available at forecast time). TFT uses a sequence-to-sequence LSTM to encode historical data, then a self-attention decoder uses this summary to predict the future. This decoder produces all future steps in a single forward pass. At a longer horizon, where the attention's signal quality (measured by attention entropy) decays as temporal context diffuses due to a longer look-back window, which may contribute to quantile crossing.

The two models thus represent *structurally* uncertainty features. Each architecture has a distinct, theoretically predictable failure mode as h increases, making them suitable for comparing how horizon-conditional miscalibration differs across different model structural features. Both models are natively implemented in GluonTS [11] with standardised dataset splits with full reproducibility.

EnbPI [1] is selected as the post-hoc conformal prediction wrapper because it relaxes time series data distributional constraints, and remains valid under non-stationary features of electricity, traffic, and retail demand data. Also, it provides a coverage guarantee without model retraining by updating calibration scores with a sliding-window residual method.

C. Evaluation Metrics

TABLE I
EVALUATION METRICS

Metric	Applies to	Measures	Formula
CRPS@h	Both	Probabilistic accuracy at h	$\int_{-\infty}^{\infty}(F(x) - \mathbf{1}[x \geq y])^2 dx$
Coverage@h	Both	Conditional coverage at level α	$\frac{1}{T}\sum_t \mathbf{1}[y_t \in \hat{C}_t]$
PIT KS@h	Both	Calibration deviation (uniform test)	$\sup_u \lvert F_n(u) - u \rvert$
Crossing@h	TFT only	Quantile ordering violations	$\frac{1}{T}\sum_t \mathbf{1}[\hat{q}_{\tau_1,t} > \hat{q}_{\tau_2,t}]$
Width@h	TFT only	Sharpness at each h	$\hat{q}_{1-\alpha/2,t} - \hat{q}_{\alpha/2,t}$
WIS@h	TFT only	CRPS approximation (K=11, corr≈1)	$\frac{1}{K+\frac{1}{2}}[\frac{1}{2}\lvert y-m \rvert + \sum_k \frac{\alpha_k}{2}\text{IS}_{\alpha_k}]$

Evaluation Metric Settings. Table I summarises the evaluation metrics and the models to which each applies.

Width@h and Crossing@h are TFT-specific metrics. DeepAR, the interval width is automatically determined by the Gaussian variance with no meaningful interpretation. For TFT, width is an independent output from separate quantile heads and can become negative when heads cross, making Width@h the main metric for the quantile-crossing hypothesis.

WIS@h is used to approximate CRPS@h for TFT as it produces quantile outputs rather than a full predictive distribution. Quantile levels $\mathcal{Q}$ follow Bracher et al. [10]'s convention: total K=11 central prediction intervals at nominal levels $1-\alpha_k$ where $\alpha_k \in \{0.02, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}$. Then, the quantiles are paired $\tau_{\text{low}} = \alpha_k/2$ and $\tau_{\text{high}} = 1-\alpha_k/2$, plus the median τ=0.50, total in $|\mathcal{Q}|$=23 heads. Bracher et al. [10] demonstrates that K=11 achieves $\text{corr}(\text{WIS}, \text{CRPS}) \approx 1.0$, making WIS@$h$ a metric comparable to CRPS@h.

The h^* tolerance is set to $\varepsilon = 0.05$ following the conventional 5% significance threshold. The horizon is classified as miscalibrated when observed Coverage@h falls more than five percentage points below the nominal level.

PIT KS@h is PIT combined with the KS (Kolmogorov–Smirnov) statistic method with the formula of PIT KS@$h = \sup_u |F_n(u) - u|$. $F_n(u)$ is the empirical CDF of $\{\text{PIT}_{t,h}\}_{t=1}^{T}$ evaluated at u where u reflects the perfectly uniform distribution. The closed-form critical value is $c_{0.05} \approx 1.36/\sqrt{T}$ where T is the time origin count, which examines whether post-EnbPI PIT KS@h falls below $c_{0.05}$. Horizons that are above the threshold are regarded as miscalibrated.

V. Data, Ethics, and Practical Considerations

A. Dataset Selection Justification

Three datasets are selected based on three criteria: (i) they should be publicly available under a permissive license, (ii) they are used in prior research, which enables comparison with

TABLE II
DATASET STRUCTURAL COMPARISON FOR h^* ANALYSIS

Dimension	Electricity	Traffic	M5
Granularity	Hourly	Hourly	Daily
Max horizon H	24 (1 day)	24 (1 day)	28 (4 weeks)
Series count (raw)	370†	963†	42,840‡
Value range	Unbounded	Bounded [0,1]	Count (zero-heavy)
Seasonality	Strong daily + weekly	Strong daily	Moderate weekly
Stationarity	Non-stationary	Near-stationary	Non-stationary
DeepAR output dist.	Gaussian	Gaussian (misfit)	Negative Binomial (count)
DeepAR time feats	Hour, weekday	Hour, weekday	Weekday, month
TFT output	Quantile ($\|\mathcal{Q}\|$=23, K=11)	Quantile ($\|\mathcal{Q}\|$=23, K=11)	Quantile ($\|\mathcal{Q}\|$=23, K=11)
TFT static	Series ID	Sensor ID	Store, Item, Cat, State
TFT known future	Hour, weekday	Hour, weekday	Weekday, month, events, SNAP, price§

†Raw UCI source counts (matching DeepAR [5]); a self-managed Band B→A pipeline reduces these to model-ready series.
‡All 12 hierarchy levels of the M5 competition [18]; bottom-level (Store×Item) = 30,490 series.
§M5 known-future inputs: promotional/holiday event indicators, state-level SNAP (Supplemental Nutrition Assistance Program) food-stamp flags, and weekly item sell prices [18].

reported baselines, and (iii) they differ along four structural features: granularity, domain, series count, and statistical properties, which enable comparing whether h^* is driven dataset-type-dependent or model-dependent features. This research includes dataset structure as one of primary dimensions of analysis as M5 Uncertainty Competition [12] reports that best model performance varies by dataset structure.

Table II summarises the structural comparison. The three datasets span two granularities (hourly vs. daily), three domains (energy, traffic, retail), and a two orders of magnitude difference across series count (370 to 42,840). The statistical properties of three datasets show differences in value range, stationarity, and seasonality patterns. All three counts in Table II refer to the size of time series in each dataset (UCI ElectricityLoadDiagrams [19]; UCI PEMS-SF [20]; the M5 Uncertainty competition archive [18]).

B. Data Readiness Assessment

Under the Lawrence Data Readiness Level (DRL) framework, all three datasets are at **Band B** with no missing values, which further require data pre-processing to be model-ready Band-A class before training. **UCI Electricity** has 15-minute granularity, which requires hourly aggregation per client time series. **UCI Traffic** has a 440-day × 963 lane sensors × 144-timestamp dimension, which requires concatenating the per-day matrices for 963 sensors with an hourly aggregated form. **M5** has bottom-level [18] time series data of 30,490. To have full dataset, aggregation of additional 11 levels is required for 42,840 time series.

UCI Electricity and **UCI Traffic** are shared through the UCI Machine Learning Repository under CC BY 4.0, which allows reuse and modification as long as the original source is credited. **M5 Walmart Sales**, released through the Kaggle M5 competition, remains under CC BY-NC, so it can only be used for non-commercial purposes. Because M5 is non-commercial, any model or result trained on it inherits the same restriction [18]. For this reason, O4 (the open-source release) is only validated on UCI Electricity and Traffic datasets.

None of the three datasets contains personal data under UK GDPR, and the chance of identifying an individual from them is negligible. Moreover, how Walmart collected internal sales data is not published, so the data's origin is only partially traceable, which is a gap under the EU AI Act Article 10 data-provenance requirement.

C. Ethical Audit and Red-Teaming

Pillar 1 – Fairness and Representation. This risk applies only to **M5 Walmart**. The dataset consists only of stores for three US states (California, Texas, Wisconsin). Also, lower-income areas may exhibit higher volatility, which can lead to systematically worse horizon-conditional calibration. If the evaluation protocol is deployed for inventory allocation, it may harm lower-income communities through consistent or vice versa, which is an indirect discrimination risk under the Equality Act 2010.

Mitigation: Disaggregate evaluation metrics such as Coverage@h and PIT KS@h by state and department. Set alarm for any store-department group whose Coverage@h breaks down from the global mean by more than ε=5%. This will mitigate a fairness concern under investigation. UCI Electricity and Traffic contain no demographic proxy variables.

Pillar 2 – Privacy and Identity. All three datasets are aggregate data, which is under the UK GDPR "motivated intruder" test, re-identification risk is almost none. Electricity records hourly consumption per meter ID, thus linking a meter ID to an individual requires the utility's non-public subscriber datasets. Traffic records lane-occupancy per PeMS sensor, which is not at the individual vehicle level. M5 records daily sales aggregates with no transaction or customer-related data, such as Customer ID. The M5 SNAP columns encode state-level food availability, which is a calendar variable, not individual-level data. Thus, a Data Protection Impact Assessment is required under UK GDPR Article 35.

Pillar 3 – Harm and Misuse. Two dual-use risks are identified.

(i) *Electricity Grid miscalibration:* an overconfident 90% interval that achieves only 60% true coverage, causing under-provisioned reserved capacity, leading to blackouts as a conse-

quence. *Mitigation:* The evaluation protocol must report whole per-horizon coverage before any operational recommendation.

(ii) *Workforce surveillance in retail field (M5):* per-store calibration breakdowns can be reused to penalise staff whose shift time overlaps with the days when the forecasts were overconfident. *Mitigation:* the protocol identifies store- and department-level breakdowns as *model* failures which is not related to staff-performance. The protocol usage is restricted to inventory and capacity planning.

Pillar 4 – Data Governance. UCI datasets carry CC BY 4.0 and M5 is CC BY-NC, so academic use is permitted but commercial deployment of M5-derived models may constitute infringement.

VI. Project Planning

The tasks of this research are managed by Trello[1] and a GitHub repository[2]. The 12-week research timeline is shown in the Gantt chart in Fig. 1.

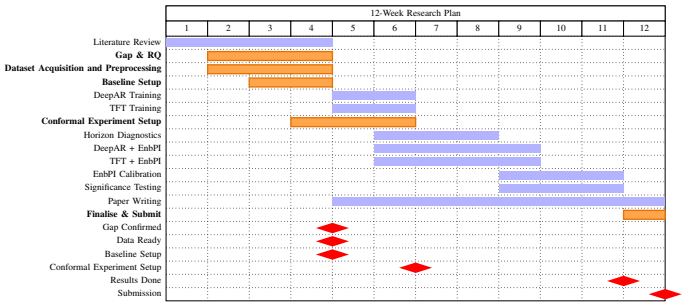


Fig. 1. 12-week project Gantt chart. (Blue: tasks. **Orange**: critical path. Red diamonds: milestones.

Key Milestones: Six milestones progress along the critical path (orange bar). *Gap Confirmed* and *Data Ready* block experimentation since no model training can begin until the research question is fixed and all three datasets are preprocessed. *Baseline Setup* sets up baseline testing code infrastructure that DeepAR and TFT are runnable on each dataset. *Conformal Experiment Setup* sets up code infrastructures that the EnbPI algorithm can be applied to DeepAR and TFT with three datasets. *Results Done* freezes all experimental outputs, leaving one full week for paper finalisation. *Submission* is the hard deadline paper submission.

Risk Assessment: Three risks are identified whithin project planning: (i) *Training convergence failure*: DeepAR and TFT may fail to converge on three datasets, which can be mitigated by using the learning-rate schedule and early-stopping convention from Salinas et al. [5] and Lim et al. [7].

(ii) *Conformal methods fail to improve coverage*: if EnbPI does not extend h^*, this will be reported as a contribution. It implies that the miscalibration at longer horizons cannot be corrected as the residuals become non-exchangeable within the calibration window.

[1]Trello board: https://trello.com/b/W2GdRUsa/dsmp-hv25324

[2]GitHub repository: https://github.com/UoB-DSMP2026/research-portfolio-hv25324

(iii) *EnbPI bootstrap retraining cost*: EnbPI [1] requires B full model re-trainings per ensemble. This can be mitigated by starting experiments using the minimum recommended B=20 [1], and parallelising the B bootstrap runs across multiple GPUs. Additionally, scheduling long training runs on cloud computing resources can be an alternative mitigation.

VII. Non-Technical Summary

Modern AI forecasting models do not just provide a single point of prediction. They provide a *range* of prediction. A power grid operator's AI might say, " Tomorrow, it is predicted with a 90% chance that the housing electricity demand will fall between 420 and 480 megawatts (MW)." With AI's prediction, the operator will then provision just enough reserve capacity to cover the maximum demand of 480MW, and also expecting that only 10% of the time AI's statement is false. Retailers also similarly make decisions of how much stock to order from outputs of their AI. Traffic authorities use AI predictions to plan signal timing for the lane on weekdays at 6 pm. In each case, the AI-stated prediction range directly leads to a real-world decision.

When that confidence of AI's prediction is well-calibrated, where a predicted 90% interval actually contains the true value 90% of the time, decision-makers can plan decisions relying on the its stated prediction. However, when a model is overconfident, where its stated 90% confidence actually delivers only 70% coverage in the real world, the consequences are tangible: chronically under-provisioned power capacity, empty shelves, and over-crowded lanes, all possibly leading to financial losses from miscalculated risk.

The problem this research addresses is that *no one is currently checking whether the accuracy of prediction from AI models degrades as their predictions look further into the future.* Previous works' evaluation tools report a single averaged accuracy across all forecast horizons. A model that is well-calibrated only for one hour ahead but increasingly unreliable for 6 days ahead can still look good in prediction performance as strong short-term performance hides long-term failure.

This project proposes a horizon-by-horizon diagnosis by measuring how accurately each model's confidence is predicted at every time step. One hour ahead, two hours ahead, all the way to 24 hours for electricity and traffic data, or all the way to 28 days later for retail sales. Where confidence accuracy is found to degrade, a statistical post-processing method is applied to restore the models' confidence. The goal is to make AI forecasting more honest, reliable, and useful for anyone whose decisions depend on knowing not just whether the prediction is going to happen, but how certain that prediction really is.

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